QUEUE AND RESOURCE OPTIMIZATION FOR PHILIPPINE EXPRESSWAY TOLLS

FINAL PROJECT FOR SIMULATION AND PERFORMANCE OPTIMIZATION

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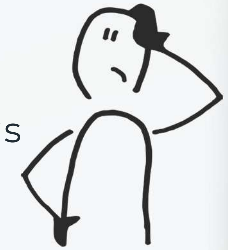
THE PROBLEM

PERSISTENT EXPRESSWAY CONGESTION

Traffic Congestion

persistent problem on Philippine expressways



Toll gate operations often contribute significantly to delays, especially during peak travel periods.

increases travel time, fuel consumption, emissions, driver frustration.

This project utilizes discrete-event simulation with ARENA software to model this complex queuing system, analyze its performance, and identify optimal configurations to reduce congestion.

MODELING THE SYSTEM: CORE ASSUMPTIONS

To create a robust simulation, the toll plaza was modeled as a classic queuing system based on the following standard assumptions:

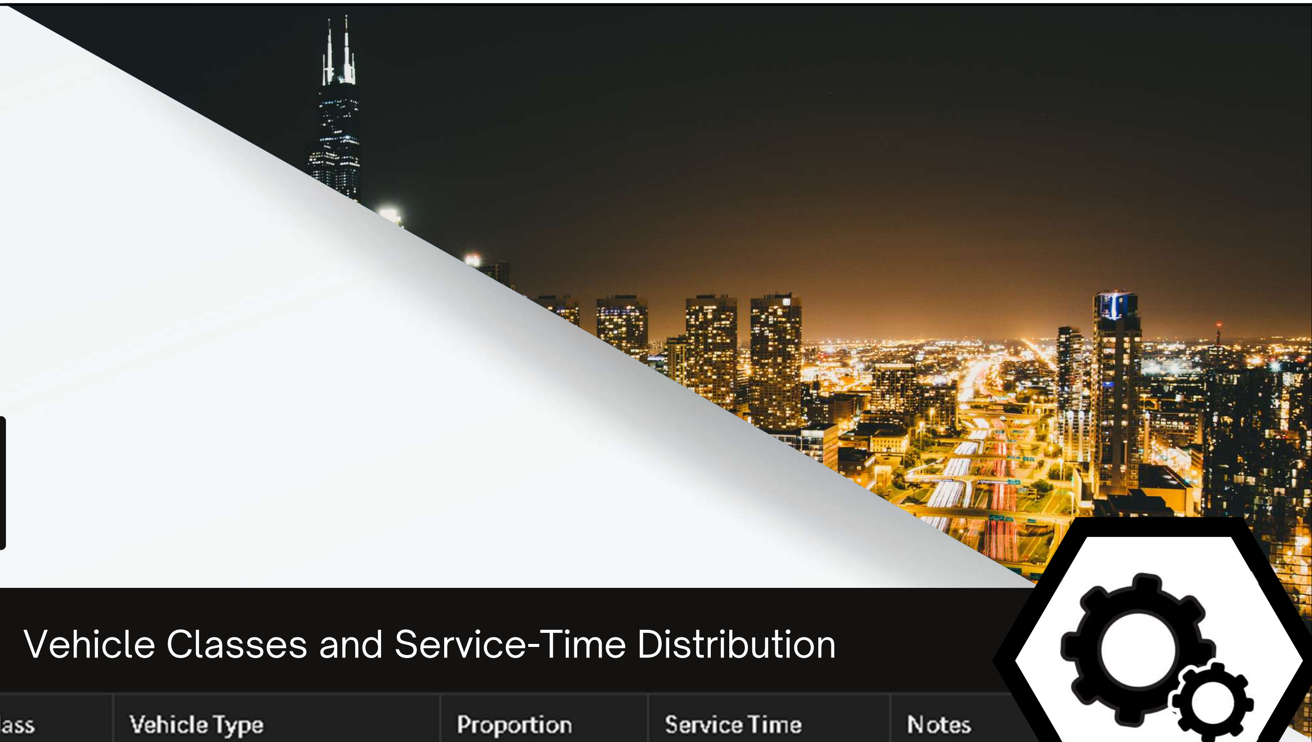
01 Poisson Process Arrivals
Vehicle arrivals follow a Poisson process generated by exponential interarrival times to model unscheduled traffic in the toll plaza.

02 Exponential Service Times
The time taken to service a vehicle is variable and follows an exponential distribution, capturing the differences in transaction times based on vehicle class.

03 First-Come, First-Served (FCFS)
Vehicles are served in the order they arrive.

04 Continuous Operation
The model assumes the system operates continuously without major equipment breakdowns disrupting the flow

BASELINE CONFIGURATION



2 cash toll collectors

Arrival Rate: $\lambda=10$ vehicles/hour

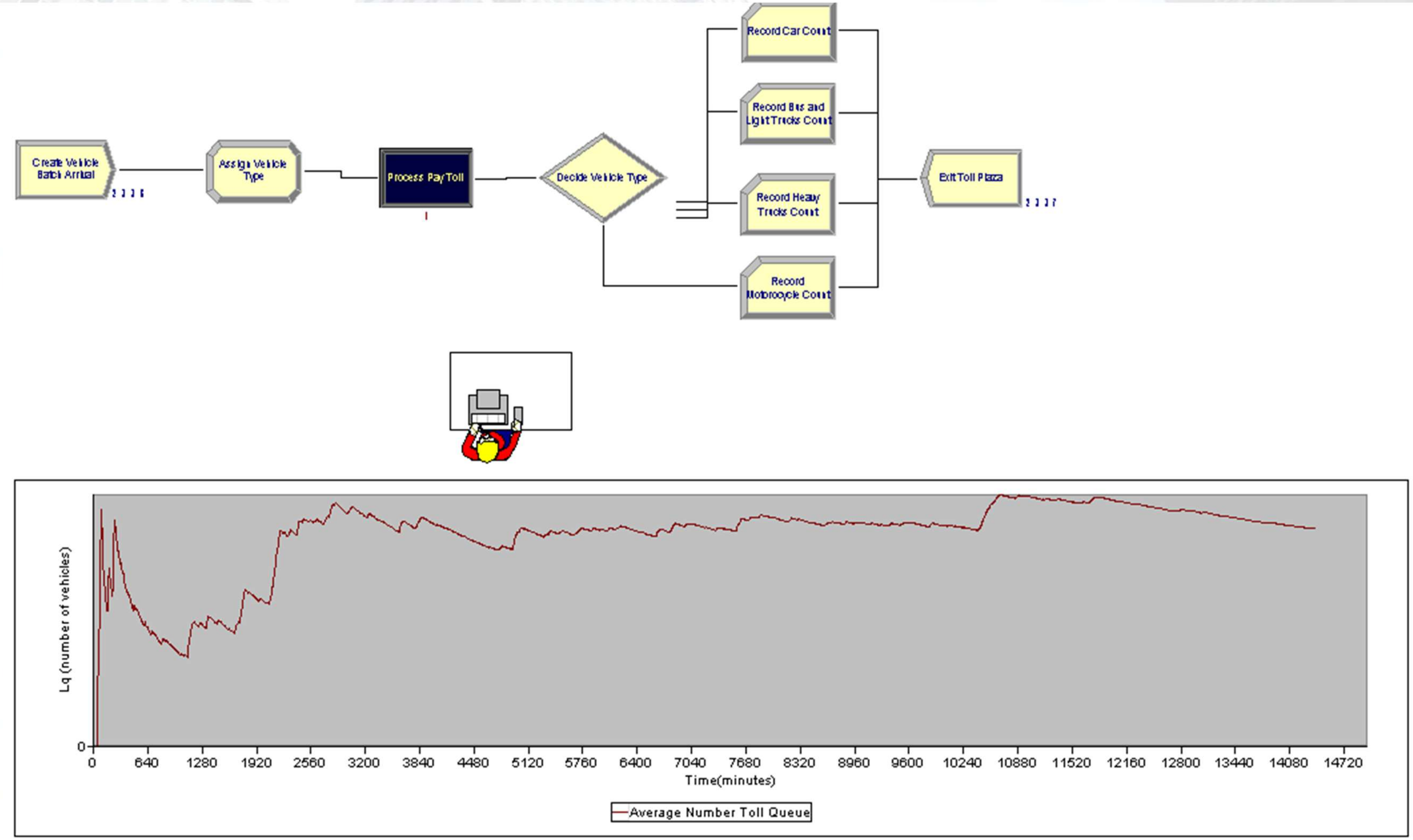
$E[T]=\lambda=0.1$ hour=6 minutes

FCFS service discipline

Vehicle Classes and Service-Time Distribution

Class	Vehicle Type	Proportion	Service Time Distribution	Notes
Class 0	Motorcycle (<400cc)	5%	Exponential, mean 1.5 min	Quick service
Class 1	Car, SUVs, Jeepneys	70%	Exponential, mean 3 min	Standard service
Class 2	Buses, Light Trucks	10%	Exponential, mean 5 min	Longer service
Class 3	Heavy Trucks	15%	Exponential, mean 6 min	Longer service

ARENA IMPLEMENTATION (BASELINE CONFIG)



SIMULATION RESULTS:

TALLY VARIABLES

Identifier	Average	Half Width	Minimum	Maximum	Observations
Process Pay Toll.TotalTimePerEntity	9.2426	1.7853	.00129	58.970	2337
Process Pay Toll.VATimePerEntity	3.5780	.13175	.00129	36.266	2337
Process Pay Toll.WaitTimePerEntity	5.6646	1.6898	.00000	55.789	2337
Vehicle.VATime	3.5780	.13175	.00129	36.266	2337
Vehicle.NVATime	.00000	.00000	.00000	.00000	2337
Vehicle.WaitTime	5.6646	1.6898	.00000	55.789	2337
Vehicle.TranTime	.00000	.00000	.00000	.00000	2337
Vehicle.OtherTime	.00000	.00000	.00000	.00000	2337
Vehicle.TotalTime	9.2426	1.7853	.00129	58.970	2337
Process Pay Toll.Queue.WaitingTime	5.6626	1.6898	.00000	55.789	2338

DISCRETE-CHANGE VARIABLES

Identifier	Average	Half Width	Minimum	Maximum	Final Value
Vehicle.WIP	1.5001	.30733	.00000	13.000	1.0000
Toll Collector.NumberBusy	.58074	.03608	.00000	1.0000	1.0000
Toll Collector.NumberScheduled	1.0000	(Insuf)	1.0000	1.0000	1.0000
Toll Collector.Utilization	.58074	.03608	.00000	1.0000	1.0000
Process Pay Toll.Queue.NumberInQueue	.91940	.27766	.00000	12.000	.00000

- **Average Toll Collector Utilization = 0.581 (58.1%)**
  - This indicates that the toll lane has sufficient capacity under baseline demand
- **Despite low arrival rate (EXPO(6)), vehicles create a queue due to the variety of service times.**
  - This indicates that service-time variability, not arrival intensity, is the primary contributor to delay
- **Possible causes of occasional congestion:**
  - Heavy trucks or buses with long service time
  - FCFS discipline (short service vehicles cannot bypass long ones)

ENHANCED MODEL

- Compound Arrival
- Time-dependent Staffing Schedules
- Multiple toll lanes(Cash and RFID)

A. Compound Arrival

Each arrival event generates a random-sized batch of vehicles, which is subsequently separated into individual entities before entering the toll booth queue.

Batch Distribution:

- 1 vehicle: 60%
- 2 vehicles: 25%
- 3 vehicles: 15%

ENHANCED MODEL

- Compound Arrival
- Time-dependent Staffing Schedules
- Multiple toll lanes(Cash and RFID)

B. Time-Dependent Staffing and Arrival Schedule

Time-Varying Vehicle Arrival Schedule

Period	Time (hrs)	Mean Interarrival Time (min)
Late Night	0–5	8.0
Morning Peak	5–9	3.0
Midday	9–16	7.0
Evening Peak	16–20	3.0
Night	20–24	9.0

Staffing Schedule:

Time (hour)	Period	Cash Collectors
0–5	Late night (Off-peak)	1
5–9	Morning peak (peak)	2
9–16	Midday (off-peak)	1
16–20	Evening peak (peak)	2
20–24	Night (off-peak)	1

ENHANCED MODEL

- Compound Arrival
- Time-dependent Dtaffing Schedules
- Multiple toll lanes(Cash and RFID)

C. Multiple Lanes

Two types of toll lanes are modeled:

1.Cash Lane

- Resource: *Cash Toll Collector*
- Capacity: *Time-dependent schedule*

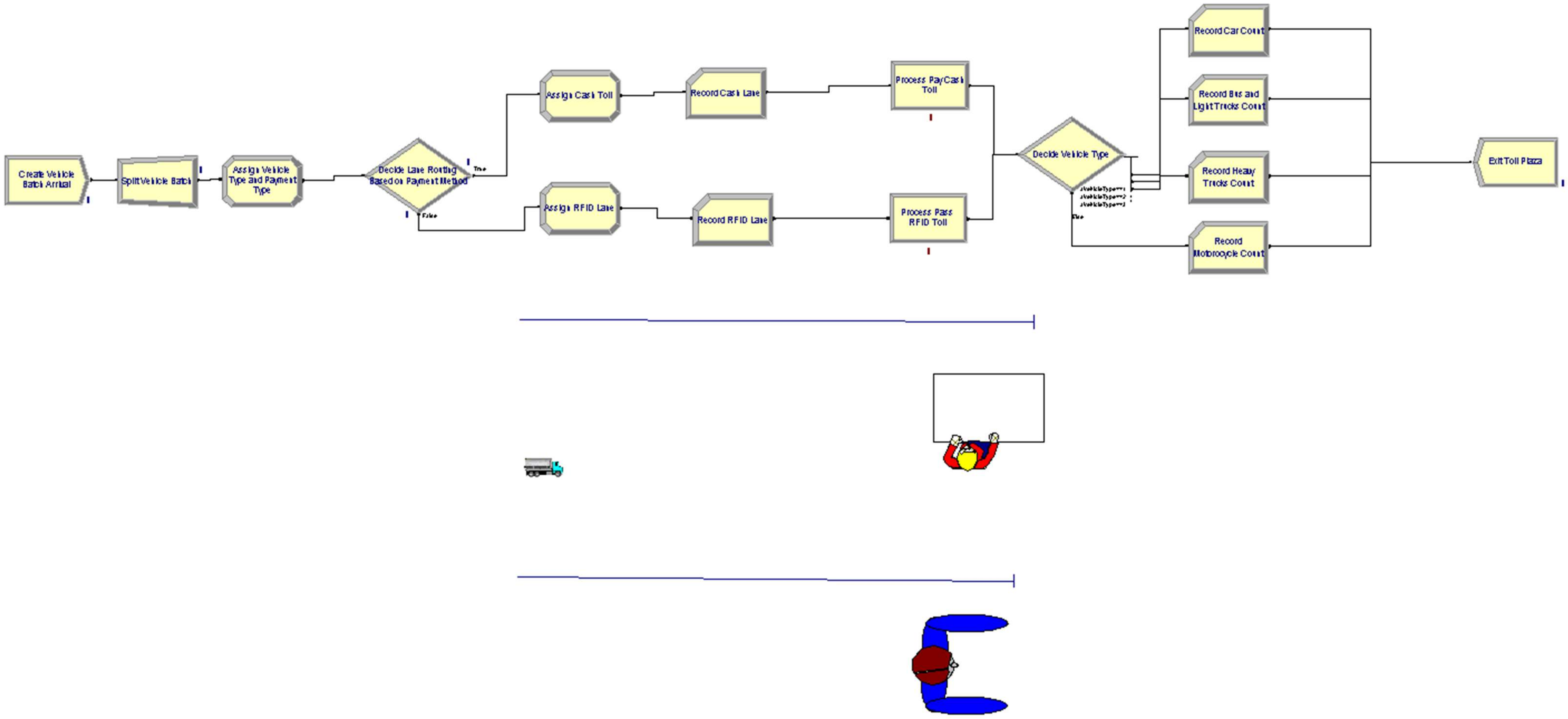
At arrival, vehicles are assigned a payment type:

- 70% Cash
- 30% RFID

2. RFID Lane

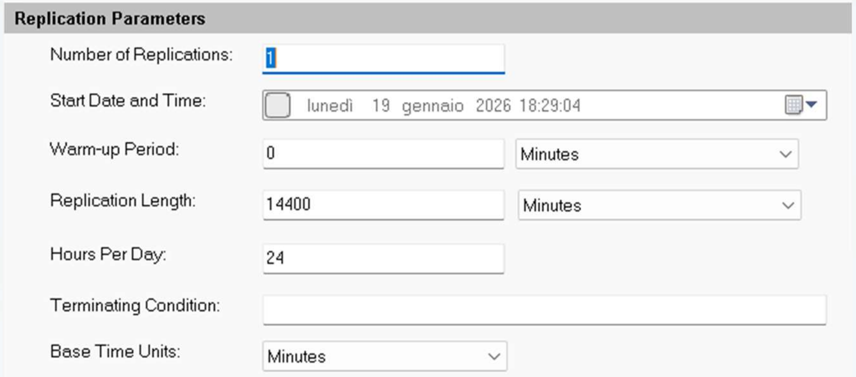
- Resource: *RFID Lane*
- Capacity: *1*
- Shorter service time than cash transactions
  - Mean Service Time ≈ 1 minute (0.0167 hrs)
  - Equivalent Service rate:
    $\mu = 1/0.01671 \approx 60$ vehicles/hour

ARENA IMPLEMENTATION

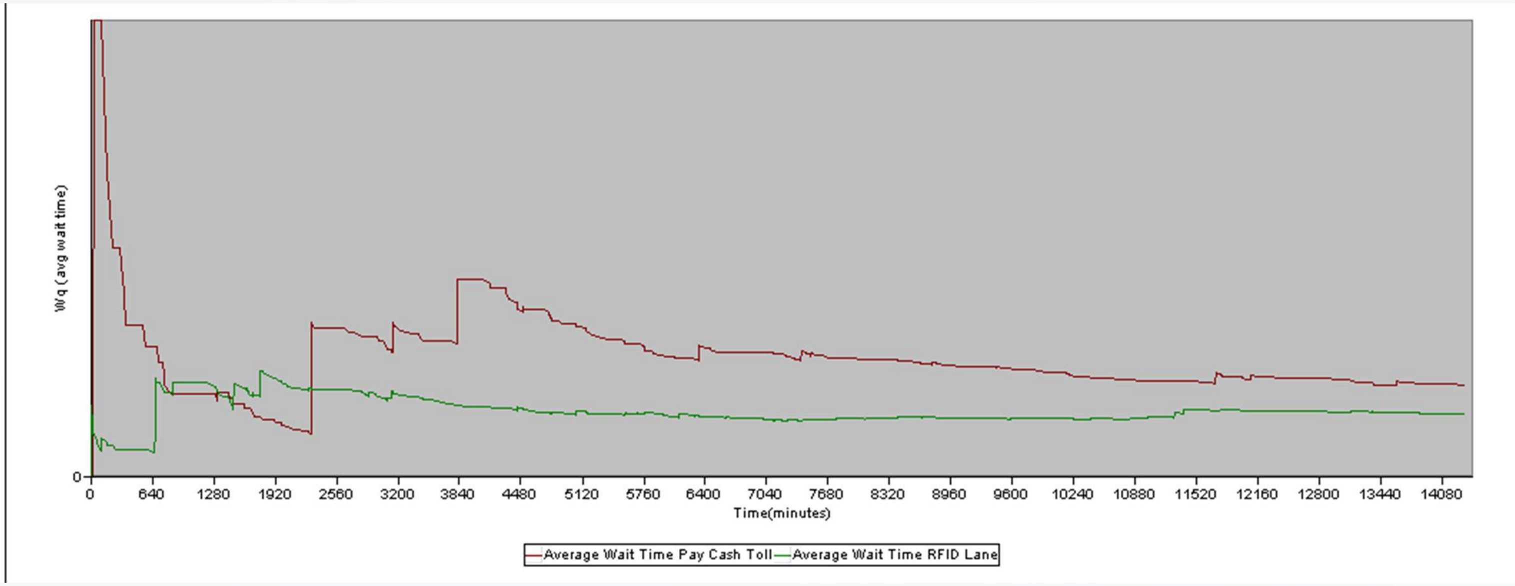


WARM UP IDENTIFICATION (ENHANCED MODEL)

Initial Replication Parameters:



Graph:

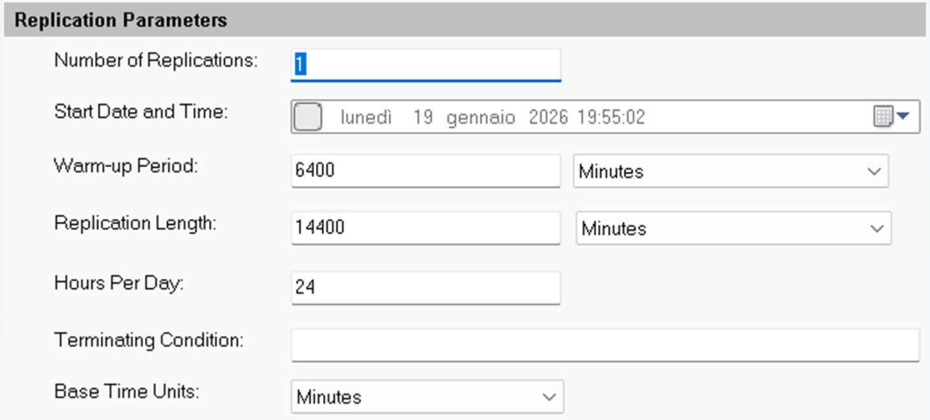


Initial Observation:

- Instability in the beginning of both curves: Cash Lane (red curve) shows very large values and sharp fluctuation, RFID (green curve) shows noticeable oscillation although much lower.
- RFID lane stabilizes earlier because of shorter service time (EXPO(0.0167)
- Reasonable visual estimate: **T_0≈6000-6400** minutes is the warm-up period. After this, Wq fluctuates around a steady mean.

FINAL EXPERIMENTAL PARAMETERS

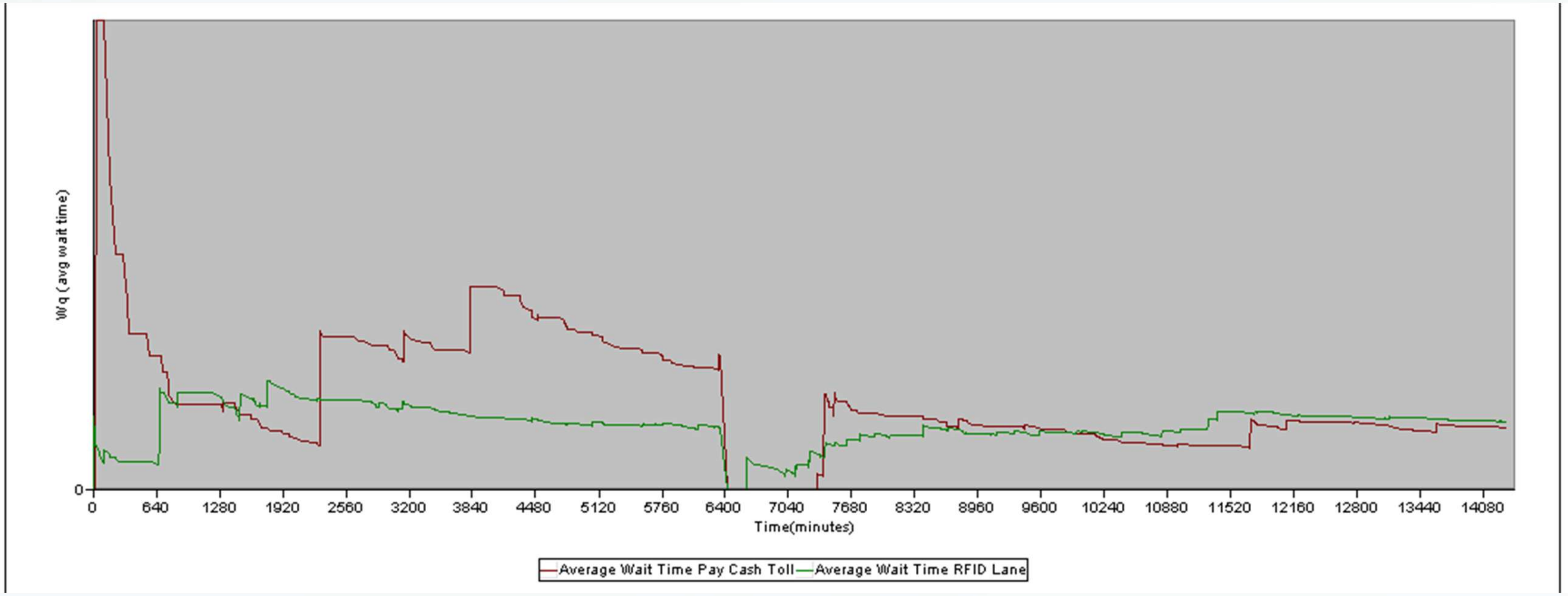
Replication Parameters (with warm-up period):



Replication Parameters

Number of Replications:	1
Start Date and Time:	lunedì 19 gennaio 2026 19:55:02
Warm-up Period:	6400 Minutes
Replication Length:	14400 Minutes
Hours Per Day:	24
Terminating Condition:	
Base Time Units:	Minutes

Graph:



RESULT OF ENHANCED MODEL

Overall system performance improvement (in comparison with the results of the baseline configuration):

Metric	Baseline	Enhanced
Avg vehicle waiting time	5.6646 min	0.31373 min
Avg total time in system	9.2426 min	1.8971 min
Avg queue length	0.9194 veh	0.00849(RFID)+ 0.00367(Cash) ≈ 0.01 veh
Max Waiting time	55.789 min	8.4933 min

- Dramatic reduction in waiting time
- Service-time variability is neutralized through electronic tolling
- RFID vehicles experience near free-flow conditions (**Avg Waiting time**: 0.32 minutes (~19 seconds), **Avg Service time**: 1.10 minutes, **Avg total time**: 1.42 min)
- Even cash users benefit significantly due to reduced competition from RFID-equipped vehicles. (**Avg Waiting Time**: 0.30 minutes, **Avg service Time**: 2.61 min, **Avg Total Time**: 2.91min)

OPTIMIZATION: WHICH CONFIGURATION IS TRULY THE BEST?

With multiple ways to improve the system, we need a statistically rigorous method to determine the most effective one. We used the **Indifference Zone (IZ) approach**, a ranking-and-selection procedure, to compare competing system designs and identify the best-performing alternative system with 95% confidence, assuming that differences smaller than a pre-specified indifference parameter are practically insignificant.

THE GOAL:

Select the system configuration that minimizes the *average vehicle waiting time*, our primary measure of congestion.

COMPETING ALTERNATIVES

Alternative A (Baseline)

1 RFID lane
2 cash collectors

Alternative B

2 RFID lanes
2 cash collectors

Alternative C

1 RFID lane
3 cash collectors

THE INDIFFERENCE-ZONE (IZ) PROCEDURE

01 Select Procedure Settings

- Number of alternative systems: **k=3**
- Probability of correct selection: **P_cs=95%**
- Indifference Zone Parameter: **δ=0.1248 mins.**
- Critical constant: **c_3,0.95=2.7101**

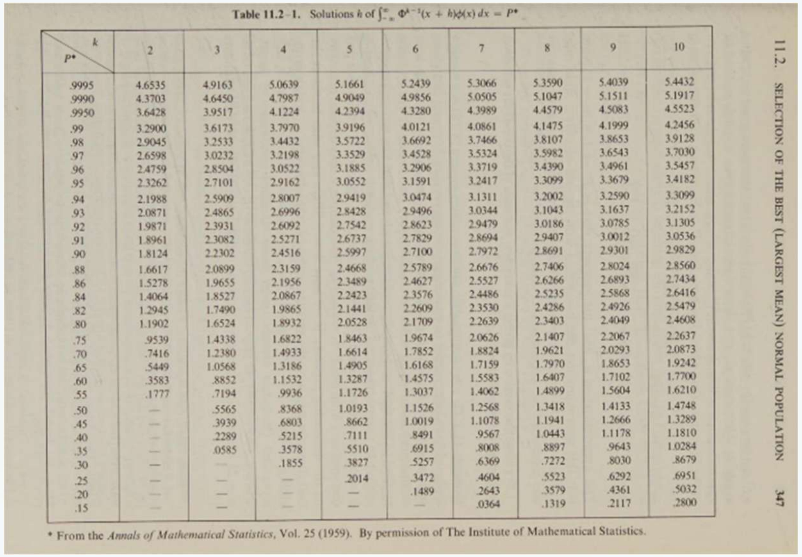


Table 11.2-1. Solutions h of $\int_{-\infty}^{\infty} \Phi^{k-1}(x+h)\phi(x)\,dx = P^*$

P^* \ k	2	3	4	5	6	7	8	9	10
.9995	4.6535	4.9163	5.0639	5.1661	5.2439	5.3066	5.3590	5.4039	5.4432
.9990	4.3703	4.6450	4.7987	4.9049	4.9856	5.0505	5.1047	5.1511	5.1917
.9950	3.6428	3.9517	4.1224	4.2394	4.3280	4.3989	4.4579	4.5083	4.5523
.99	3.2900	3.6173	3.7970	3.9196	4.0121	4.0861	4.1475	4.1999	4.2456
.98	2.9045	3.2533	3.4432	3.5722	3.6692	3.7466	3.8107	3.8653	3.9128
.97	2.6598	3.0232	3.2198	3.3529	3.4528	3.5324	3.5982	3.6543	3.7030
.96	2.4759	2.8504	3.0522	3.1885	3.2906	3.3719	3.4390	3.4961	3.5457
.95	2.3262	2.7101	2.9162	3.0552	3.1591	3.2417	3.3099	3.3679	3.4182
.94	2.1988	2.5909	2.8007	2.9419	3.0474	3.1311	3.2002	3.2590	3.3099
.93	2.0871	2.4865	2.6996	2.8428	2.9496	3.0344	3.1043	3.1637	3.2152
.92	1.9871	2.3931	2.6092	2.7542	2.8623	2.9479	3.0186	3.0785	3.1305
.91	1.8961	2.3082	2.5271	2.6737	2.7829	2.8694	2.9407	3.0012	3.0536
.90	1.8124	2.2302	2.4516	2.5997	2.7100	2.7972	2.8691	2.9301	2.9829
.88	1.6617	2.0899	2.3159	2.4668	2.5789	2.6676	2.7406	2.8024	2.8560
.86	1.5278	1.9655	2.1956	2.3489	2.4627	2.5527	2.6266	2.6893	2.7434
.84	1.4064	1.8527	2.0867	2.2423	2.3576	2.4486	2.5235	2.5868	2.6416
.82	1.2945	1.7490	1.9865	2.1441	2.2609	2.3530	2.4286	2.4926	2.5479
.80	1.1902	1.6524	1.8932	2.0528	2.1709	2.2639	2.3403	2.4049	2.4608
.75	.9539	1.4338	1.6822	1.8463	1.9674	2.0626	2.1407	2.2067	2.2637
.70	.7416	1.2380	1.4933	1.6614	1.7852	1.8824	1.9621	2.0293	2.0873
.65	.5449	1.0568	1.3186	1.4905	1.6168	1.7159	1.7970	1.8653	1.9242
.60	.3583	.8852	1.1532	1.3287	1.4575	1.5583	1.6407	1.7102	1.7700
.55	.1777	.7194	.9936	1.1726	1.3037	1.4062	1.4899	1.5604	1.6210
.50	—	.5565	.8368	1.0193	1.1526	1.2568	1.3418	1.4133	1.4748
.45	—	.3939	.6803	.8662	1.0019	1.1078	1.1941	1.2666	1.3289
.40	—	.2289	.5215	.7111	.8491	.9567	1.0443	1.1178	1.1810
.35	—	.0585	.3578	.5510	.6915	.8008	.8897	.9643	1.0284
.30	—	—	.1855	.3827	.5257	.6369	.7272	.8030	.8679
.25	—	—	—	.2014	.3472	.4604	.5523	.6292	.6951
.20	—	—	—	—	.1489	.2643	.3579	.4361	.5032
.15	—	—	—	—	—	.0364	.1319	.2117	.2800

* From the *Annals of Mathematical Statistics*, Vol. 25 (1959). By permission of The Institute of Mathematical Statistics.

Based on the table from Bechhofer (1954)

THE INDIFFERENCE-ZONE (IZ) PROCEDURE

02 Determine Sample Size N

$$N = \left\lceil \left(\sigma \cdot c_{k,P^*} / \delta\right)^2 \right\rceil$$

To obtain the variance of the performance measure, I conducted an initial run with n0=10 replications to estimate reliably before sampling.

n=	10		
	Initial Run		
i= / n=	1	2	3
1	0,27423	0,10792	0,25695
2	2,7887	1,5622	2,7862
3	1,0053	0,19572	0,99409
4	4,1337	3,4989	4,522
5	0.85708	0.68223	0,85161
6	0,93479	0,47368	0,96119
7	1,3006	0,78305	1,286
8	0,83015	0,6313	0,83015
9	1,0682	0,34518	1,0959
10	0,47949	0,29913	0,47949
Xi	1,28152	0,789708	1,40636
Si^2	1,54101	1,156014	1,65669
oled variance=		1,451236	

$$N = \frac{(2.7101)^2(1.4512)^2}{(0.1248)^2}$$

$$= 123.9043 \approx 124$$

THE INDIFFERENCE-ZONE (IZ) PROCEDURE

03 Run N Replications for each System

Replication Parameters

Number of Replications:	124
Start Date and Time:	mercoledì 21 gennaio 2026 17:01:03
Warm-up Period:	6400 Minutes
Replication Length:	1000 Minutes
Hours Per Day:	24
Terminating Condition:	
Base Time Units:	Minutes

Output Summary for each Alternative System:

ARENA Simulation Results
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Output Summary for 124 Replications

Project: Unnamed Project — Run execution date : 1/21/2026
Analyst: suboljhenalyn3@gmail.com — Model revision date: 1/21/2026

OUTPUTS

Identifier	Average	Half-width	Minimum	Maximum	# Replications
Process Pay Cash Toll Number In	44.927	5.3859	4.0000	157.00	124
Process Pay Cash Toll Number Out	44.604	5.2505	4.0000	136.00	124
Process Pass RFID Toll Accum VA Time	105.58	12.547	16.885	331.78	124
Process Pass RFID Toll Number Out	104.90	12.580	13.000	340.00	124
Process Pass RFID Toll Number In	104.97	12.589	13.000	341.00	124
Process Pass RFID Toll Accum Wait Time	92.959	29.061	2.8759	1103.9	124
Process Pay Cash Toll Accum Wait Time	283.19	214.10	.00000	10199.	124
Process Pay Cash Toll Accum VA Time	158.20	19.610	5.8588	531.77	124
Vehicle.NumberIn	149.90	17.821	17.000	498.00	124
Vehicle.NumberOut	149.50	17.675	17.000	468.00	124
RFID Lane.NumberSeized	104.95	12.589	13.000	341.00	124
RFID Lane.ScheduledUtilization	.10562	.01255	.01689	.33251	124
Cash Toll Collector.NumberSeized	44.701	5.2570	4.0000	136.00	124
Cash Toll Collector.ScheduledUtilization	.12430	.01543	.00458	.41567	124
System.NumberOut	149.50	17.675	17.000	468.00	124

Simulation run time: 8.03 minutes.
Simulation run complete.

ARENA Simulation Results
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Output Summary for 124 Replications

Project: Unnamed Project — Run execution date : 1/21/2026
Analyst: suboljhenalyn3@gmail.com — Model revision date: 1/21/2026

OUTPUTS

Identifier	Average	Half-width	Minimum	Maximum	# Replications
Process Pay Cash Toll Number In	43.516	5.0706	3.0000	133.00	124
Process Pay Cash Toll Number Out	43.403	5.0509	3.0000	129.00	124
Process Pass RFID Toll Accum VA Time	105.00	11.999	14.430	302.95	124
Process Pass RFID Toll Number Out	103.78	11.729	10.000	324.00	124
Process Pass RFID Toll Number In	103.82	11.729	11.000	324.00	124
Process Pass RFID Toll Accum Wait Time	9.1991	2.3471	.00000	73.763	124
Process Pay Cash Toll Accum Wait Time	197.22	99.241	.00000	4413.2	124
Process Pay Cash Toll Accum VA Time	154.42	18.479	5.6088	437.17	124
Vehicle.NumberIn	147.33	16.643	14.000	450.00	124
Vehicle.NumberOut	147.18	16.625	13.000	450.00	124
RFID Lane.NumberSeized	103.82	11.729	11.000	324.00	124
RFID Lane.ScheduledUtilization	.05255	.00600	.00744	.15148	124
Cash Toll Collector.NumberSeized	43.500	5.0616	3.0000	131.00	124
Cash Toll Collector.ScheduledUtilization	.12115	.01450	.00438	.35681	124
System.NumberOut	147.18	16.625	13.000	450.00	124

Simulation run time: 8.17 minutes.
Simulation run complete.

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Output Summary for 124 Replications

Project: Unnamed Project — Run execution date : 1/21/2026
Analyst: suboljhenalyn3@gmail.com — Model revision date: 1/21/2026

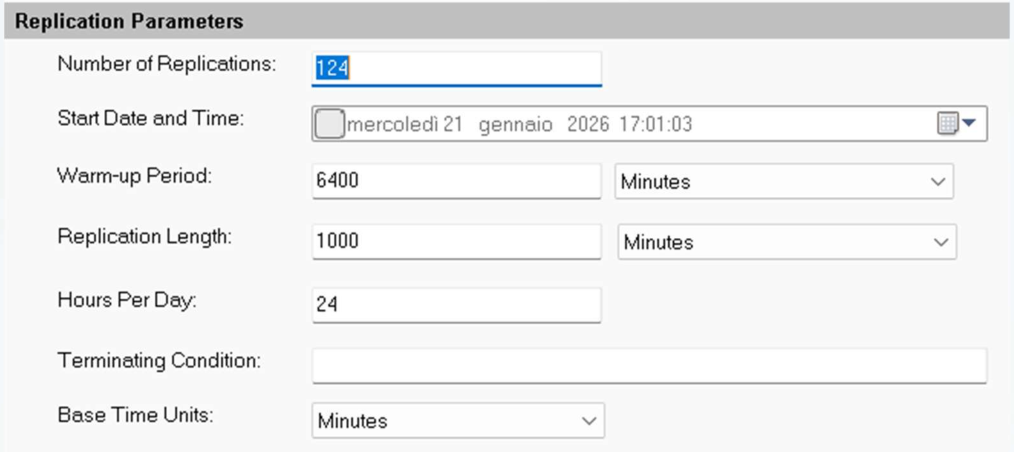
OUTPUTS

Identifier	Average	Half-width	Minimum	Maximum	# Replications
Process Pay Cash Toll Number In	45.048	5.3800	4.0000	156.00	124
Process Pay Cash Toll Number Out	44.830	5.3138	4.0000	142.00	124
Process Pass RFID Toll Accum VA Time	105.41	12.473	16.885	323.31	124
Process Pass RFID Toll Number Out	104.88	12.545	13.000	336.00	124
Process Pass RFID Toll Number In	105.00	12.553	13.000	336.00	124
Process Pass RFID Toll Accum Wait Time	92.834	29.039	2.8759	1103.9	124
Process Pay Cash Toll Accum Wait Time	287.10	219.92	.00000	9797.9	124
Process Pay Cash Toll Accum VA Time	158.90	19.787	5.8588	547.43	124
Vehicle.NumberIn	150.04	17.775	17.000	492.00	124
Vehicle.NumberOut	149.71	17.704	17.000	478.00	124
RFID Lane.NumberSeized	104.96	12.552	13.000	336.00	124
RFID Lane.ScheduledUtilization	.10548	.01248	.01689	.32332	124
Cash Toll Collector.NumberSeized	44.959	5.3243	4.0000	145.00	124
Cash Toll Collector.ScheduledUtilization	.10244	.01277	.00376	.36643	124
System.NumberOut	149.71	17.704	17.000	478.00	124

Simulation run time: 9.98 minutes.
Simulation run complete.

THE INDIFFERENCE-ZONE (IZ) PROCEDURE

03 Run N Replications for each System



Results of the performance measure (Vehicle.WaitTime) are organized in this file

iz-excel.pdf

THE INDIFFERENCE-ZONE (IZ) PROCEDURE

04 Compute Sample Mean (124 Replications)

Xi	1,0163184	0,786468	1,346601
Si^2	1,0844984	2,253881	8,942662

05 Select the Best System

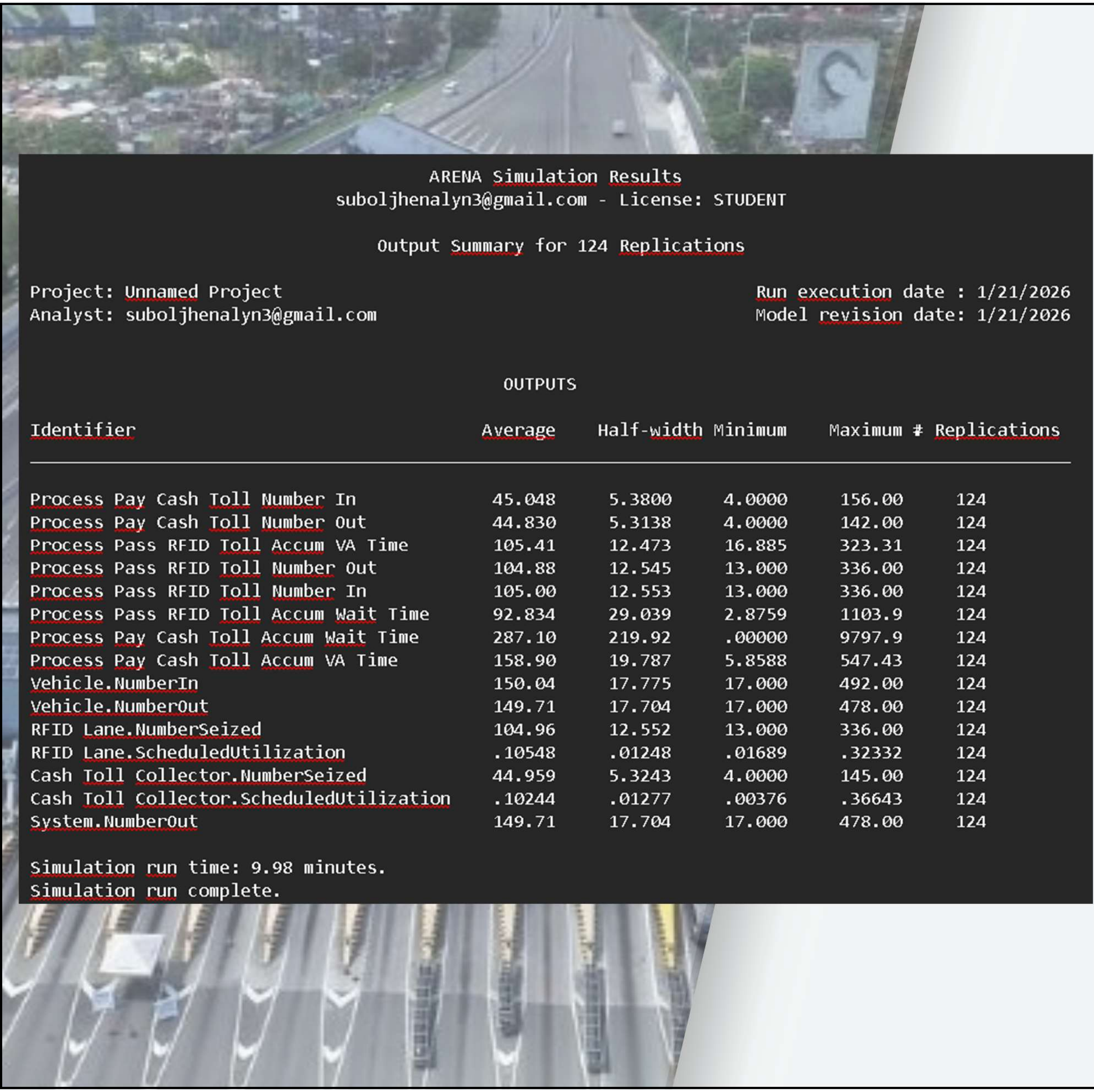
Because we are looking for the *minimum* average waiting time, we transform the averages Xi to negative

$$i^* = argmin\{-1.02, -0.79, -1.35\}$$

Thus,

$$i^* = 3$$

Alternative System C yields the minimum average waiting time.



ARENA Simulation Results
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Output Summary for 124 Replications

Project: Unnamed Project
Analyst: suboljhenalyn3@gmail.com

Run execution date : 1/21/2026
Model revision date: 1/21/2026

OUTPUTS

Identifier	Average	Half-width	Minimum	Maximum	# Replications
Process Pay Cash Toll Number In	45.048	5.3800	4.0000	156.00	124
Process Pay Cash Toll Number Out	44.830	5.3138	4.0000	142.00	124
Process Pass RFID Toll Accum VA Time	105.41	12.473	16.885	323.31	124
Process Pass RFID Toll Number Out	104.88	12.545	13.000	336.00	124
Process Pass RFID Toll Number In	105.00	12.553	13.000	336.00	124
Process Pass RFID Toll Accum Wait Time	92.834	29.039	2.8759	1103.9	124
Process Pay Cash Toll Accum Wait Time	287.10	219.92	.00000	9797.9	124
Process Pay Cash Toll Accum VA Time	158.90	19.787	5.8588	547.43	124
Vehicle.NumberIn	150.04	17.775	17.000	492.00	124
Vehicle.NumberOut	149.71	17.704	17.000	478.00	124
RFID Lane.NumberSeized	104.96	12.552	13.000	336.00	124
RFID Lane.ScheduledUtilization	.10548	.01248	.01689	.32332	124
Cash Toll Collector.NumberSeized	44.959	5.3243	4.0000	145.00	124
Cash Toll Collector.ScheduledUtilization	.10244	.01277	.00376	.36643	124
System.NumberOut	149.71	17.704	17.000	478.00	124

Simulation run time: 9.98 minutes.
Simulation run complete.

RESULT

Alternative C shows stable, low-congestion performance across 124 replications.

The Cash Toll processed 45 vehicles on average (95% CI: 39.63-50.28) with high wait time variability (accumulative wait 287.1 min total). Meanwhile, RFID Lane handles 105 vehicles efficiently (95% CI: 92.41-117.5) with low utilization (10.55%) and minimal accumulated wait (92.8 min total).

Despite the accumulated wait variability (Cash: 287.1 min, HW=219 min; RFID: 92.8 min, HW=29 min), a per-vehicle analysis confirms System C's preference zone superiority through the Vehicle.WaitTime:

System C(X3=1.35) vs System A (X1=1.02):
X3-X1=0.33 > δ=0.12 ✓ *Preference Zone*

System C(X3=1.35) vs System B (X2=0.79):
X3-X2=0.56 >> δ=0.12 ✓ *Strong Preference*

CONCLUSION

The results indicate that increasing the number of cash toll collectors during peak hours provides the most significant reduction in congestion under the *assumed traffic composition*. While additional RFID lanes improve performance, congestion in the cash lane remains the dominant contributor to system delay.

Through the application of the **Indifference Zone (IZ) ranking-and-selection procedure**, this selection is validated with the specified confidence level, accounting for stochastic variability across replications.